

CXR Bootcamp Labs Compendium
Connections, live adjustments, and UI evidence
`cxr-ops-lab` + rehearsal `cxr-ui`

CXR Ops Lab

2026-05-31

Abstract

Single printable index for implemented software labs: how each tool connects to **Claim Studio (:3000)** and **rehearsal dev (:8251)**, files added, fixes applied during bootcamp, and verified UI screenshots. Individual deep-dive PDFs live alongside this compendium in **docs/**.

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Chapter 1

How bootcamp wiring works (master view)

1.1 Two CXR entry points

URL	What it is	Which labs touch it
:3000 /claim-studio	Compose cxr-ui (product dashboard)	SW.12 logs only
:8251 /claim-studio	Rehearsal npm dev	SW.11 OTel traces

1.2 Pattern: every SW lab uses the same wiring recipe

Each elective (SW.12–SW.18) is a **standalone Compose file** under `cxr-ops-lab/`:

Step	Artifact	Example (SW.17 Vault)
1	<code>compose*.yaml</code>	<code>compose.vault.yaml</code> → container + ports
2	<code>lab/<tool>/</code>	<code>seed-cxr-secrets.sh</code> , <code>cxr-secret-map.json</code>
3	<code>scripts/NN*-up.sh</code>	<code>20-vault-up.sh</code> wraps <code>compose up</code>
4	<code>scripts/NN*-smoke.sh</code>	Health check + golden-path verify
5	<code>evidence/SW*.md</code> + PNG	Checklist + UI screenshot
6	<code>docs/CXR*-LAB-MANUAL.pdf</code>	This compendium + per-lab PDF

1.3 Connection matrix (what talks to what)

Lab	Data flow	Wired to :3000 / :8251?
SW.11 OTel	:8251 app → :4318 → Jaeger :16686	Yes (:8251)
SW.12 ELK	:3000 container logs → Filebeat → Kibana	Yes (:3000 logs)
SW.13 Kafka	script → :9092 topic → UI :8082	No
SW.14 Redis	script → :6379 → Insight :5540	No
SW.15 GraphQL	:4000 → :4001/:4002 subgraphs	No
SW.16 gRPC	grpcui :8090 → :50051 ClaimAnalysis	No
SW.17 Vault	seed script → KV <code>secret/cxr/*</code> → UI :8200	No (future K8)
SW.18 Langfuse	<code>send-trace.mjs</code> → :3100 UI	No (future judge/LLM)

Design choice: electives run *beside* Claim Studio so you learn each tool without rewriting the app every time.

Chapter 2

Product dashboard vs observability UIs

URL	Role	Wired in lab?
:3000 /claim-studio	Main CXR dashboard (Compose)	Log source (SW.12)
:8251 /claim-studio	Rehearsal dev + OTel traces	Trace source (SW.11)
:16686	Jaeger — traces	SW.11
:5601	Kibana — logs	SW.12
:5540	Redis Insight — cache keys	SW.14
:4000	Apollo Sandbox — GraphQL	SW.15
:8082	Kafka UI — topics/messages	SW.13
:8090	grpcui — gRPC invoke	SW.16
:8200	Vault — secrets UI	SW.17
:3100	Langfuse — LLM traces	SW.18
:3001	Grafana — metrics	SW.9–10

Chapter 3

Lab PDF index

SW	PDF	UI
SW.11 OTel	CXR-OTEL-LAB-MANUAL.pdf	Jaeger :16686
SW.12 ELK	CXR-ELK-LAB-MANUAL.pdf	Kibana :5601
SW.13 Kafka	CXR-KAFKA-LAB-MANUAL.pdf	Kafka UI :8082
SW.14 Redis	CXR-REDIS-LAB-MANUAL.pdf	Redis Insight :5540
SW.15 GraphQL	CXR-GRAPHQL-LAB-MANUAL.pdf	Apollo :4000
SW.16 gRPC	CXR-GRPC-LAB-MANUAL.pdf	grpcui :8090
SW.17 Vault	CXR-VAULT-LAB-MANUAL.pdf	Vault :8200
SW.18 Langfuse	CXR-LANGFUSE-LAB-MANUAL.pdf	Langfuse :3100
Deploy	CXR-K8-DEPLOYMENT-MANUAL.pdf	K8 / Argo / :8081

Build all: `./scripts/build-all-lab-pdfs.sh`

Chapter 4

SW.11 OpenTelemetry — connection + fixes + UI

Path: :8251 Claim Studio → OTLP :4318 → Jaeger :16686.

Live adjustments: Next.js 16 searchParams; instrumentation split; Resource API; serverExternalPackage stop systemd :8251 before OTEL_* dev.

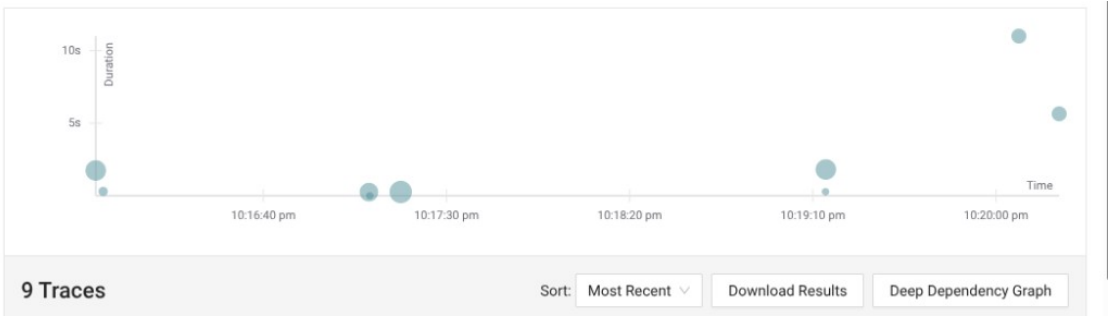


Figure 4.1: Jaeger Search (SW.11).



Figure 4.2: POST analyze waterfall (SW.11).

Chapter 5

SW.12 ELK — connection + fixes + UI

Path: :3000 cxr-ui logs → Filebeat → ES :9200 → Kibana :5601.

Live adjustments: cxr.bootcamp.logs label; Kibana Explore/Data Views path; time picker vs histogram; not :8251 logs.

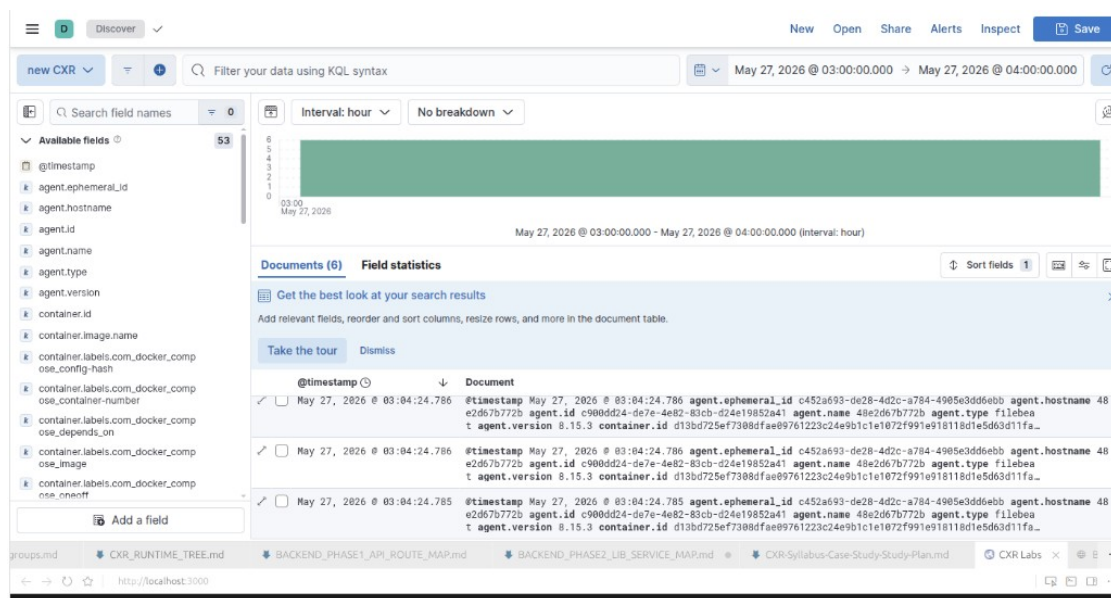


Figure 5.1: Kibana Discover (SW.12).

Chapter 6

SW.14 Redis — connection + fixes + UI

Path: cache-aside script → Redis :6379 → Redis Insight :5540.

Live adjustments: Insight service added; connect `redis://redis:6379`; container name filter; TTL/DEL timing.

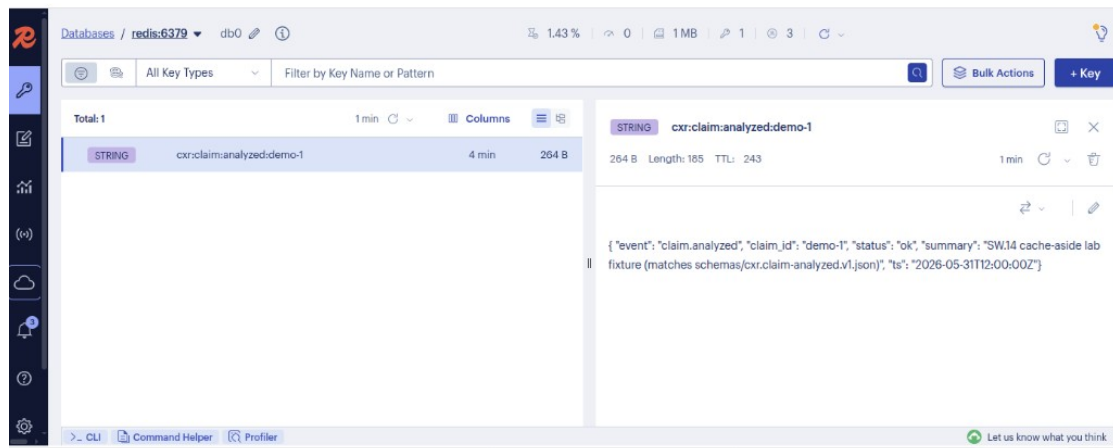


Figure 6.1: Redis Insight key view (SW.14).

Chapter 7

SW.15 GraphQL — connection + fixes + UI

Path: Browser :4000 → gateway → claims :4001 + policies :4002.

Live adjustments: parse() for federation SDL; subgraph healthchecks before gateway.

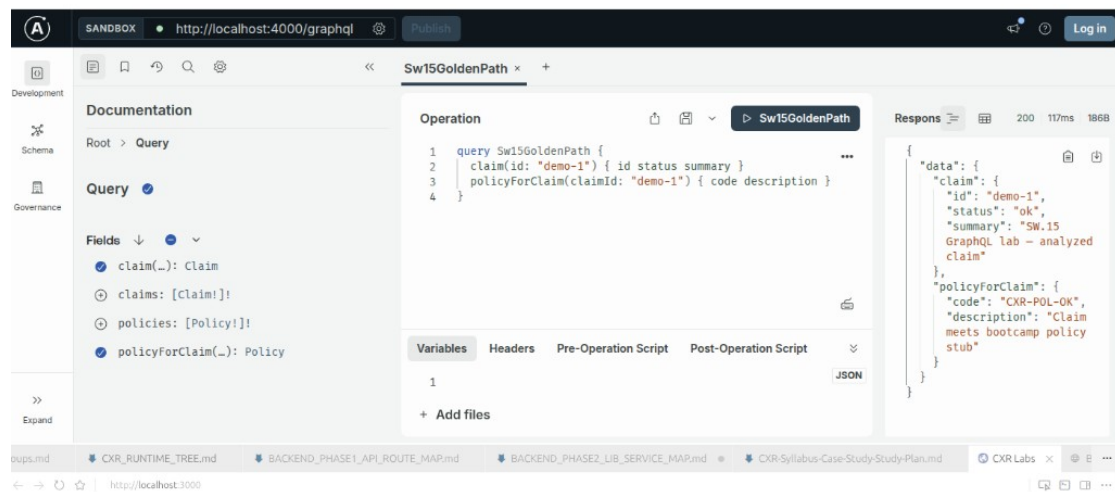


Figure 7.1: Apollo Sandbox Sw15GoldenPath (SW.15).

Chapter 8

SW.13 Kafka (UI at :8082)

Path: `kafka-produce-sample.sh` → topic `cxr.claims.events` → Kafka UI.

No archived screenshot yet — capture Messages tab in Kafka UI for optional `SW13-kafka-ui-*.png`.

Chapter 9

SW.16 gRPC — connection + UI

Path: grpcui :8090 → ClaimAnalysis :50051 → mock demo-1.

Live adjustments: @grpc/reflection on server (grpc-ui crash without it); grpc-ui restart in 19-grpc-up.sh.

Service name: »

Method name: ◀

Request Form Raw Request Response History

Request Metadata

Name	Value

+ Add item

Request Data

cxr.v1.AnalyzeClaimRequest

claim_id string ☐

content string ☐

Request Timeout

seconds

Invoke

Figure 9.1: grpcui `AnalyzeClaim` form (SW.16).

Chapter 10

SW.17 Vault — full wiring + UI

Chain: `20-vault-up.sh` → `compose.vault.yaml` → Vault :8200 → `seed-cxr-secrets.sh` (docker exec vault kv put) → `secret/cxr/analyzer`.

Login: Token only — `cxr-bootcamp-root`. UI: Secrets → `secret/cxr/analyzer`.

Not wired: Claim Studio still uses `.env`; future = External Secrets on K8 :8081.

See `CXR-VAULT-LAB-MANUAL.pdf` for step-by-step diagram.

Chapter 11

SW.18 Langfuse — wiring

Chain: 21-langfuse-up.sh → compose.langfuse.yaml (v2 + Postgres) → :3100 UI → user API keys in lab/langfuse/keys.env → send-trace.mjs → trace cxr-sw18-golden-path.

Not wired: judge_claim.py / doc-reasoning routes do not send traces yet.

Chapter 12

What is not wired (by design)

- Redis, GraphQL, Kafka, gRPC, Vault, Langfuse — not called from Claim Studio API routes.
- ELK — ships :3000 Docker logs only (not :8251 terminal dev).
- Optional: Compose :3000 OTel via `compose.otel-link.yaml` (separate from :8251 golden path).